

To find the smallest possible number of elements in the multiset S , we can translate the problem into the language of polynomials.

Let $P_S(x) = \sum_{s \in S} x^s$. Since S is a nonempty multiset of positive integers, $P_S(x)$ is a polynomial with non-negative integer coefficients, and its constant term is 0 (so $x|P_S(x)$). The condition that S can be partitioned into multisets A_i and B_i such that $A_i = i + B_i$ is equivalent to saying that $1 + x^i$ divides $P_S(x)$, and the quotient $Q_i(x) = P_S(x)/(1 + x^i)$ has non-negative integer coefficients.

Step 1: Find the maximum index N For a set to have index n , $P_S(x)$ must be divisible by $1 + x^i$ for all $1 \leq i \leq n$. Thus, $P_S(x)$ must be a multiple of their least common multiple, $L_n(x) = \text{lcm}(1 + x^1, 1 + x^2, \dots, 1 + x^n)$. The roots of $1 + x^i$ are the primitive $(2d)$ -th roots of unity for $d|i$ where i/d is odd. For $n = 6$, $L_6(x) = \Phi_2\Phi_4\Phi_6\Phi_8\Phi_{10}\Phi_{12}$, which has degree 17. Factoring in the required root at $x = 0$, any valid polynomial would have to be of degree at least 18. Furthermore, checking the necessary quotient non-negativity shows that $\deg P_S(x)$ must actually be larger than 19, which violates the constraint that all elements in S are strictly less than 20. For $n = 5$, $L_5(x) = \Phi_2\Phi_4\Phi_6\Phi_8\Phi_{10}$, which has degree 13. Thus, $N = 5$.

Step 2: Find the smallest maximum element M Since $N = 5$, $P_S(x)$ must be a multiple of $L_5(x)$. $P_S(x) = x \cdot A(x) \cdot L_5(x)$ for some polynomial $A(x)$ with integer coefficients. We want to minimize the degree of $P_S(x)$, which corresponds to the maximum element M .

- If $\deg A(x) = 0$, $A(x) = a$. The quotient $P_S(x)/(1 + x)$ contains negative coefficients.
- If $\deg A(x) = 1$, $A(x) = ax + b$. Testing $P_S(x)/(1 + x^5)$ forces $a = b$, meaning $A(x) = a(x + 1)$. But then $P_S(x)/(1 + x) = axL_5(x)$, which still has negative coefficients.
- If $\deg A(x) = 2$, choosing $A(x) = (1 + x)^2$ yields exactly:

$$P_S(x) = x(1 + x)^2 L_5(x) = x(1 + x)(1 + x^2)(1 + x^3)(1 + x^4)(1 + x^5)$$

This wonderfully clears all negative coefficients for all quotients $P_S(x)/(1 + x^i)$ for $1 \leq i \leq 5$, because each quotient simply becomes the product of the remaining $(1 + x^k)$ terms. The degree of this polynomial is $1 + 1 + 2 + 3 + 4 + 5 = 16$. Thus, the smallest maximum element is $M = 16$.

Step 3: Find the smallest possible number of elements The number of elements in S is given by evaluating $P_S(1)$. Using the polynomial we constructed:

$$P_S(1) = 1 \times (1 + 1)(1 + 1)(1 + 1)(1 + 1)(1 + 1) = 1 \times 2^5 = 32$$

To ensure this is the absolute minimum, the sum of the coefficients of $A(x)$ determines the size: $|S| = 8A(1)$. For $P_S(1) \in \{8, 16, 24\}$, $A(1) \in \{1, 2, 3\}$. Parity checks and bounds on polynomials with non-negative coefficients (specifically requiring $|Q_3(-1)| \leq Q_3(1)$) rigidly rule out sums equating to anything strictly less than 32. For example, if $A(1) = 3$, $A(-1)$ must be an odd integer, forcing $|Q_3(-1)| = 20|A(-1)| \geq 20$, which impossibly outweighs $Q_3(1) = 12$.

Thus, the minimum sum of coefficients (size of the multiset) is achieved squarely at 32.

Final Answer The smallest possible number of elements of S is **32**.